

Advancing Quantum Process Tomography Through Quantum Compilation

Huynh Le Dan Linh, Vu Tuan Hai, Le Bin Ho

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Main work

1.1. Background

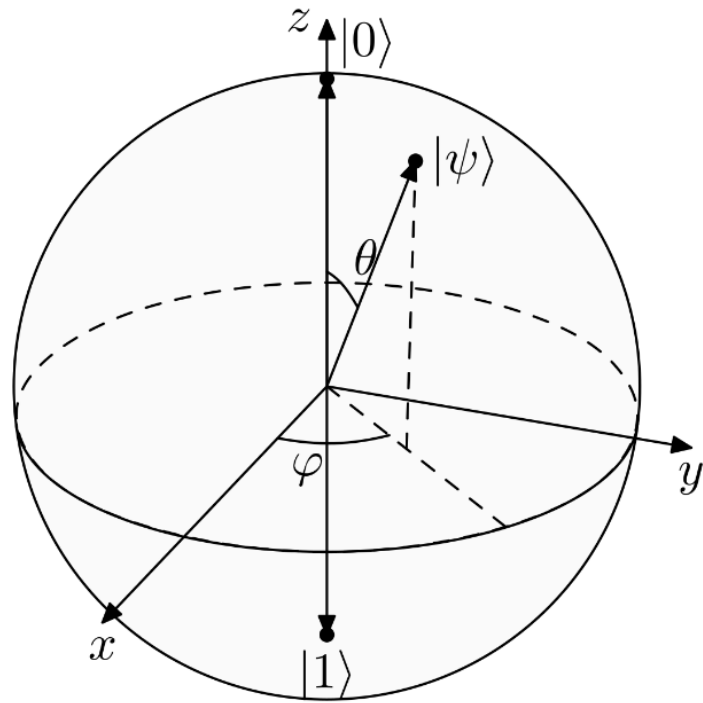


Figure 1.1. Representation of a qubit on Bloch sphere

Qubit

Unlike a classical bit (0 or 1), a qubit can be in a superposition of both states.

- **Vector representation**

$$, \text{ with } |\alpha|^2 + |\beta|^2 = 1$$

- **Bloch sphere representation**

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\varphi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

$\theta \in [0, \pi]$: polar angle

$\varphi \in [0, 2\pi)$: azimuthal angle

1.1. Background

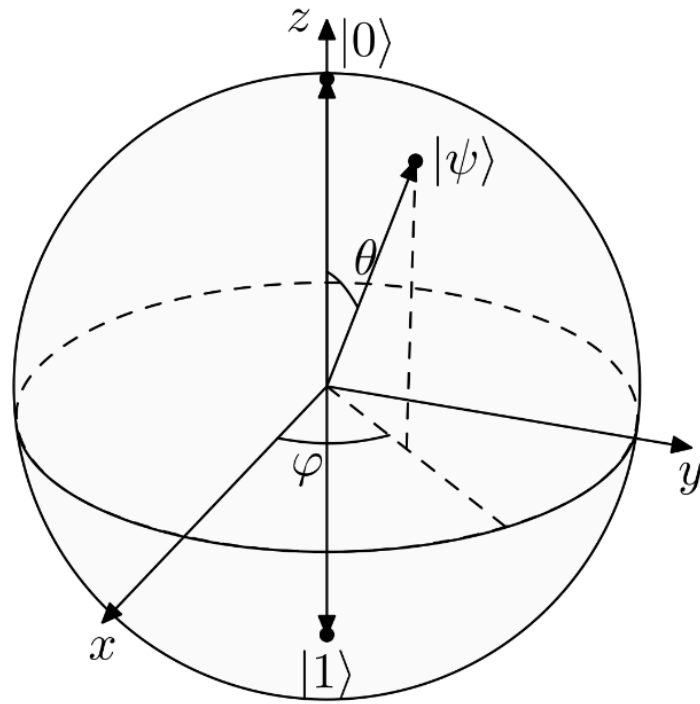


Figure 1.1. Representation of a qubit on Bloch sphere

Qubit

Unlike a classical bit (0 or 1), a qubit can be in a superposition of both states.

- Matrix representation

$$\rho = |\psi\rangle\langle\psi| = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{bmatrix}$$

$\rho_{\{00\}} = |\alpha|^2$: probability of measuring

: probability of measuring

: quantum coherence (phase information)

$\rho_{\{10\}} = \alpha^*\beta$: complex conjugate of coherence

- Diagonal terms give probabilities
- Off-diagonal terms encode quantum coherence.

1.1. Background

Quantum Process

A quantum process (quantum channel) describes state evolution: $\rho \rightarrow \rho'$

Kraus Representation

$$\rho' = \sum_i K_i \rho K_i^\dagger, \text{ with } \sum_i K_i^\dagger K_i = I$$

- K_i : Kraus operators
- Describes noise and open quantum systems
- Physical interpretation: probabilistic quantum operations

Matrix Representation

- : Choi matrix of the process
- Provides a one-to-one correspondence between quantum processes and matrices

1.1. Background

- A major challenge in reconstructing quantum states is that it is limited by the **Heisenberg uncertainty principle and measurement-induced state collapse [1]**
- Quantum Process Tomography (QPT) is a technique used to characterize unknown quantum processes and benchmark quantum devices. However, traditional QPT methods scale exponentially and are impractical for large quantum systems. [2] [3]

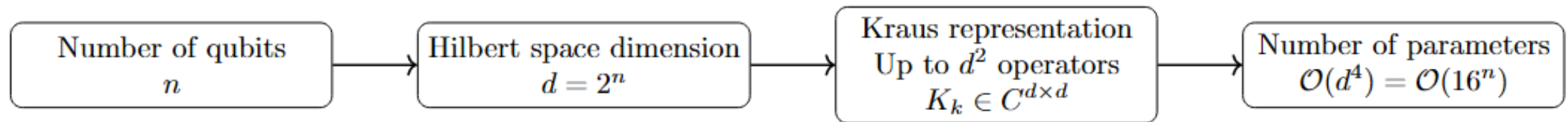


Figure 1.2. Exponential parameter scaling in Quantum Process Tomography (Kraus-Based)

[1] G. Mauro D'Ariano, Matteo G. A. Paris, Massimiliano F. Sacchi, "Quantum Tomography," Advances in Imaging and Electron Physics, vol. 128, pp. 205-308, 2003.

[2] M. Mohseni, A. T. Rezakhani, D. A. Lidar, "Quantum-process tomography: Resource analysis of different strategies," Physical Review A, vol. 77, issue 3, 2008.

[3] J.B. Altepeter, E.R. Jeffrey, P.G. Kwiat, "Photonic state tomography," trong Advances In Atomic, Molecular, and Optical Physics, vol. 52, Elsevier, 2005, pp. 105-159.

1.1. Background

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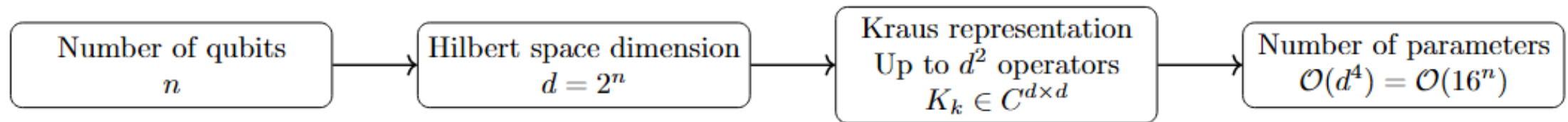


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[3] J.B. Altepeter, E.R. Jeffrey, P.G. Kwiat, "Photonic state tomography," trong Advances In Atomic, Molecular, and Optical Physics, vol. 52, Elsevier, 2005, pp. 105-159.

1.2. Research Objectives and Deliverables

- The project proposes an efficient quantum state tomography method that reduces complexity from $O(2^n)$ to $O(\log n)$, where n is the number of qubits.
- The core algorithm, **Compilation-based Quantum Process Tomography (CQPT)**, aims to minimize computational time and resources by machine learning and compilation techniques [4].

Deliverables

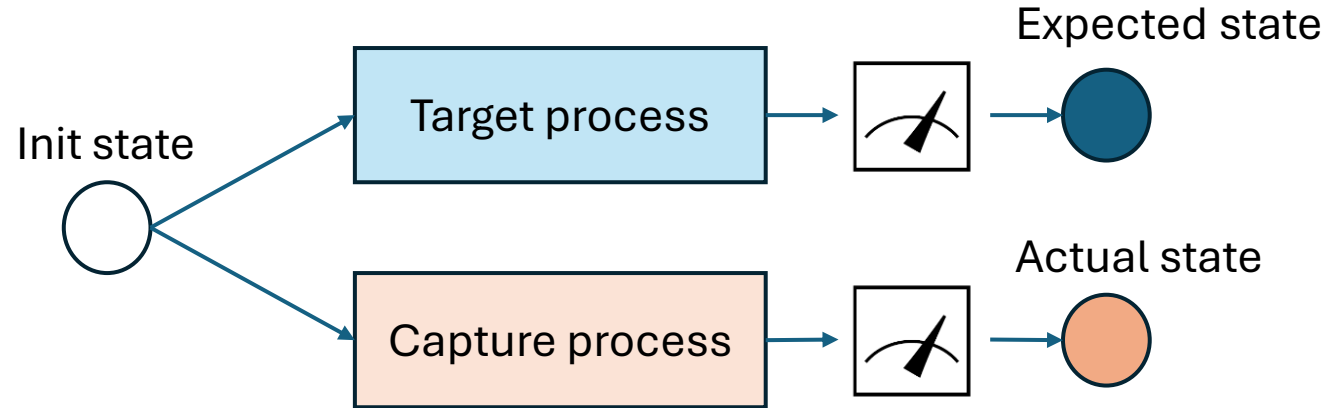
- A complete theoretical framework
- Validated performance demonstrated through experiments on various quantum processes, scaling with the number of qubits.
- A software package that integrates the full CQPT workflow.

[4] Vu Tuan Hai, Le Bin Ho, “Universal compilation for quantum state preparation and tomography,” 2022. [Online]. Available: <https://arxiv.org/abs/2204.11635>.

1.2. Research Objectives and Deliverables

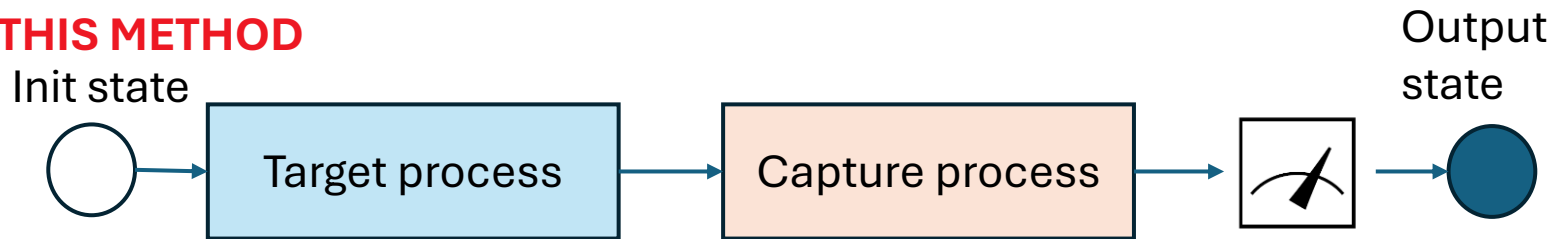
Input	A list of training states ρ (matrix) A quantum process $\mathcal{E}$ (matrix/ function)
Output	Kraus operators/ matrix k that captured the effect of $\mathcal{E}$
Evaluation	Compilation trace fidelity between output states $\rho_{\mathcal{E}}$ and ρ_k (achieving by apply original $\mathcal{E}$ and captured k on test set ρ_{test}) ~ 1.0

EXISTING METHOD



Actual state $\sim$ Expected state
 $\Rightarrow$ **Capture process $\sim$ Target process**

THIS METHOD



Output state $\sim$ Init state
 $\Rightarrow$ **Capture process $\sim$ Inverse of Target process**
 $\Rightarrow$ **Inverse of Capture process $\sim$ Target process**

1.3. Scope of Project

Experimental scope

- Theoretical and numerical experiments are conducted via simulation:
 - Input and output quantum states are represented as density matrices.
 - Quantum processes are simulated using Kraus operators and complex matrices.
 - Simulation is performed through matrix multiplication and numerical computation.

Functional scope

- Initialization of numerical data for CQPT, including:
 - Input states
 - Initial Kraus operators, initial complex matrices
 - Support for generating large sets of random states, operators, and matrices
- CQPT algorithm execution using:
 - Complex matrix modeling
 - Kraus operator modeling
- Validation of output on self-generated test states.

2.1. Recent Quantum Process Tomography approaches

Table 2.1. My research’s contribution.

Aspect	Gradient-descent QPT [5]	Riemann Gradient-descent QPT [6]	Proposed CQPT (This Work)
Core Idea	Optimize Kraus operators via gradient descent on manifold	Gradient descent on Riemannian manifold to accelerate convergence	Single-shot inverse gate measurement
Physical Constraints	CPTP preserved	CPTP preserved	CPTP preserved; evaluated with single-shot measurement
Measurements / Scalability	Large measurement data; limited by qubit number	Faster convergence; still limited by qubit number	Reduced measurements; scalable to multi-qubit systems
Strengths	Direct optimization on Kraus; preserves physical constraints	Faster optimization; maintains physical constraints	Single-shot evaluation reduces experiments; scalable

[5] Ahmed S., Quijandría F., Kockum A. F., “Gradient-descent quantum process tomography,” Phys. Rev. Lett., 2023.

[6] Volya D., Nikitin A., Mishra P., “Fast quantum process tomography via riemannian gradient descent,” 2024.

2.2. Universal Compilation technique

Quantum Channel

- A quantum channel $\mathcal{E}$ is a CPTP map representing an unknown quantum process [4]:

$$\mathcal{E}(\rho) = \sum_l K_l \rho K_l^\dagger, \quad \sum_l K_l^\dagger K_l = I$$

where K_l are **Kraus operators**.

- Alternatively, $\mathcal{E}$ can be represented by the Choi matrix $J_\mathcal{E}$ via the Choi–Jamiołkowski isomorphism:

$$J_\mathcal{E} = \sum_{\{i,j\}} |i\rangle\langle j| \otimes \mathcal{E}(|i\rangle\langle j|)$$

with $|i\rangle$ as an orthonormal basis.

- For an (N)-qubit system:

$$K_l \in \mathcal{C}^{2^N \times 2^N}, \quad J_\mathcal{E} \in \mathcal{C}^{4^N \times 4^N}$$

[4] Vu Tuan Hai, Le Bin Ho, “Universal compilation for quantum state preparation and tomography,” 2022. [Online]. Available: <https://arxiv.org/abs/2204.11635>.

2.2. Universal Compilation technique

Quantum Compilation and CQPT

- **Quantum compilation:** transform a target unitary U into a trainable unitary $V(\theta)$ parametrized by θ [7]. Universal quantum compilation ensures **any unitary can be approximated with a parameterized circuit** [8]:

$$V(\theta) \approx U \quad \text{via a quantum circuit ansatz}$$

- **CQPT (Compilation-based QPT):** uses two circuits U and $V(\theta)$, modeled via **Kraus operators or complex matrices**, optimized so that:

$$V^\dagger(\theta)U \approx Ie^{i\phi}, \quad \forall \phi$$

[7] Kentaro Heya, Yasunari Suzuki, Yasunobu Nakamura, Keisuke Fujii, “Variational quantum gate optimization,” 2018.

[8] Vu Tuan Hai, Nguyen Tan Viet, Jesus Urbaneja, Nguyen Vu Linh, Lan Nguyen Tran, Le Bin Ho, “Multi-target quantum compilation algorithm,”

2.3. Quantum Machine Learning Techniques

Numerical simulation approach

- **Approach:** simulate quantum channels entirely on classical computers using quantum channels.
 - Compute exact gradients via analytic or automatic differentiation, avoiding noisy parameter-shift techniques.
 - Use classical optimizers (e.g. ADAM) to optimize parameters.
- **Advantages:**
 - Full control over training process
 - Can adjust circuit structure and verify correctness
 - Stable convergence even for small or noisy gradients

2.3. Quantum Machine Learning Techniques

Hybrid Quantum-Classical approach

- **Approach:** combine parameterized quantum circuits with classical optimizers [9] [10].

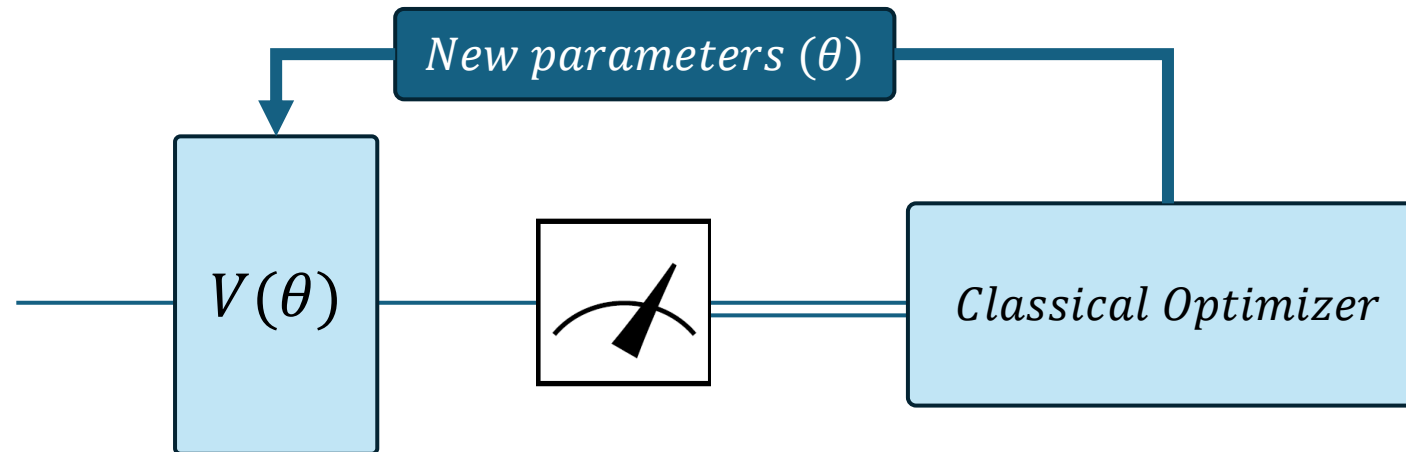


Figure 2.1. Hybrid Machine learning workflow

- **Advantages:**
 - Adapts to device noise
 - Avoids classical simulation of large matrices for many qubits

Note: This thesis focuses on classical simulation; hybrid execution is planned for future development.

[9] J. Preskill, "Quantum Computing in the NISQ era and beyond," Quantum, vol. 2, p. 79, 2018.

[10] Maria Schuld, Francesco Petruccione, "Supervised Learning with Quantum Computers", Springer, 2018.

3.1. CQPT Theoretical Framework

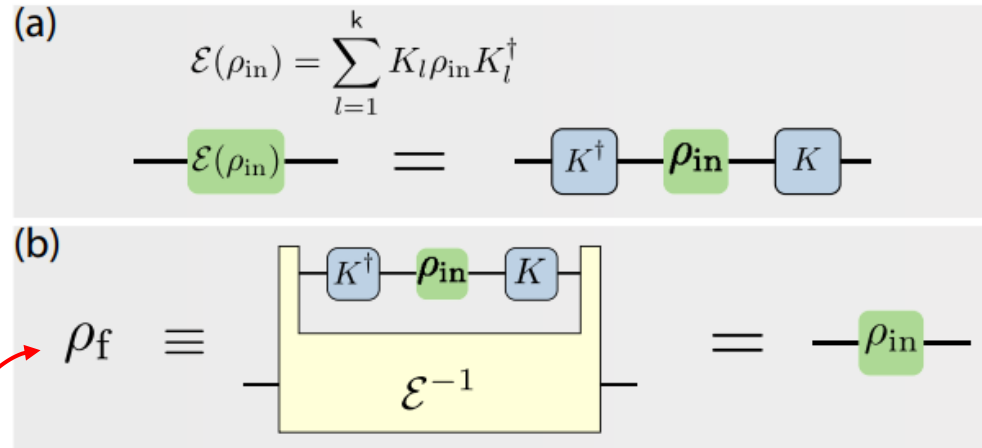


Figure 3.1. (a) Quantum channel representation with Kraus operators. (b) Visualization of Theorem 1.

Theorem 1. Training a set of Kraus operators $\mathbf{K}$ represents a quantum process $\mathcal{E}$, if the following condition holds

$$\rho_f \equiv \mathcal{E}^{-1} \left(\sum_{l=1}^k K_l \rho_{\text{in}} K_l^\dagger \right) = \rho_{\text{in}}$$

So $\mathbf{K}$ is a representation of $\mathcal{E}$

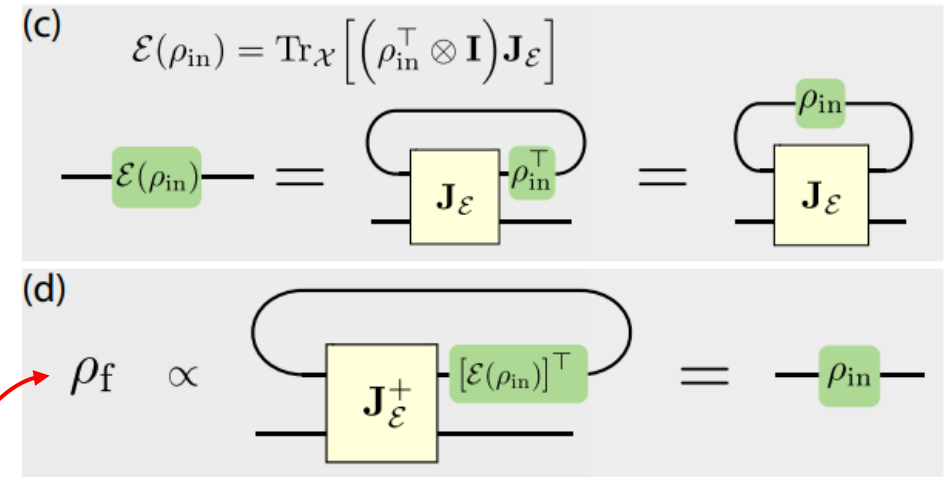


Figure 3.2. (c) Quantum channel representation with Choi matrix. (d) Visualization of Theorem 2.

Theorem 2. A trainable Choi matrix $\mathbf{J}_{\mathcal{E}} \in \mathcal{C}^{4^N \times 4^N}$ is positive semi-definite and satisfies $\text{Tr}_{\mathcal{Y}}(\mathbf{J}_{\mathcal{E}}) = \mathbf{I}$ represents a quantum process $\mathcal{E}$ if:

$$\rho_f \propto \text{Tr}_{\mathcal{X}} \left[\left(\mathcal{E}(\rho_{\text{in}})^\top \otimes \mathbf{I} \right) \mathbf{J}_{\mathcal{E}}^+ \right] = \rho_{\text{in}}$$

3.2. Algorithm on Numerical Simulation

Kraus-based CQPT

Apply Kraus operators to input states, compute cost with target, and update parameters via Riemannian ADAM until convergence.

Theorem 1. Training a set of Kraus operators $\mathbf{K}$ represents a quantum process $\mathcal{E}$, if the following condition holds

$$\rho_f \equiv \mathcal{E}^{-1} \left(\sum_{l=1}^k K_l \rho_{in} K_l^\dagger \right) = \rho_{in}$$

So $\mathbf{K}$ is a representation of $\mathcal{E}$

Algorithm 3.1. The implementation of Kraus-based CQPT

Algorithm 1: CQPT with Kraus using Riemannian Gradient + ADAM

Input: Initial Kraus operators K_0 , training states $\{\rho_{in}^{(i)}\}_{i=1}^N$, learning rate α , ADAM parameters $\beta_1, \beta_2, \epsilon$, maximum iterations T

Output: Optimized Kraus operators K

Normalize K_0 such that $\sum_l K_l^\dagger K_l = I$;

Initialize $K \leftarrow K_0$, $m \leftarrow 0$, $v \leftarrow 0$;

for $t = 1$ to T do

$C \leftarrow 0$;

for $i = 1$ to N do

$\rho_{out}^{(i)} \leftarrow \sum_l K_l \rho_{in}^{(i)} K_l^\dagger$;

$\rho_{out_inv}^{(i)} \leftarrow \mathcal{E}^{-1}(\rho_{out}^{(i)})$;

$C_i \leftarrow \text{ComputeCost}(\rho_{in}^{(i)}, \rho_{out_inv}^{(i)})$;

$C \leftarrow C + C_i$;

$C(K) \leftarrow \text{avg}(C)$;

$G \leftarrow \nabla C(K)$;

$G_R \leftarrow G - K \cdot \text{Sym}(K^\dagger G)$;

$m \leftarrow \beta_1 m + (1 - \beta_1) G_R$;

$v \leftarrow \beta_2 v + (1 - \beta_2)(G_R \odot G_R)$;

$\hat{m} \leftarrow m / (1 - \beta_1^t)$;

$\hat{v} \leftarrow v / (1 - \beta_2^t)$;

$K \leftarrow K - \alpha \cdot \hat{m} / (\sqrt{\hat{v}} + \epsilon)$;

$K \leftarrow K \cdot (K^\dagger K)^{-1/2}$;

return K

Apply process
Compute cost
& gradient

ADAM

Update Kraus

3.2. Algorithm on Numerical Simulation

Choi-based CQPT

Apply Choi operator to input states, compute cost with target, and update parameters via Riemannian ADAM until convergence.

Theorem 2. A trainable Choi matrix

$J_{\mathcal{E}} \in \mathcal{C}^{4^N \times 4^N}$ is positive semi-definite and satisfies $\text{Tr}_Y(J_{\mathcal{E}}) = \mathbf{I}$ represents a quantum process $\mathcal{E}$ if:

$$\rho_f \propto \text{Tr}_X[(\mathcal{E}(\rho_{\text{in}})^{\top} \otimes \mathbf{I})J_{\mathcal{E}}^+] = \rho_{\text{in}}$$

Algorithm 3.2. The implementation of Choi-based CQPT

Algorithm 2: CQPT with Choi Matrix using ADAM and CPTP Projection

Input: Initial Choi matrix $J_{\mathcal{E}}^{(0)}$, training states $\{\rho_{\text{in}}^{(i)}\}_{i=1}^N$, learning rate α , ADAM parameters $\beta_1, \beta_2, \epsilon$, maximum iterations T

Output: Optimized Choi matrix $J_{\mathcal{E}}$

Project $J_{\mathcal{E}}^{(0)}$ onto CPTP set;

Initialize $J_{\mathcal{E}} \leftarrow J_{\mathcal{E}}^{(0)}$, $m \leftarrow 0$, $v \leftarrow 0$;

for $t = 1$ to T do

$C \leftarrow 0$;

for $i = 1$ to N do

$\rho_{\text{out}}^{(i)} \leftarrow \mathcal{E}_{J_{\mathcal{E}}}(\rho_{\text{in}}^{(i)})$;

$\rho_{\text{out_inv}}^{(i)} \leftarrow \mathcal{E}_{J_{\mathcal{E}}}^{-1}(\rho_{\text{out}}^{(i)})$;

$C_i \leftarrow \text{ComputeCost}(\rho_{\text{in}}^{(i)}, \rho_{\text{out_inv}}^{(i)})$;

$C \leftarrow C + C_i$;

$C(J_{\mathcal{E}}) \leftarrow \text{avg}(C)$;

$G \leftarrow \nabla C(J_{\mathcal{E}})$;

$m \leftarrow \beta_1 m + (1 - \beta_1)G$;

$v \leftarrow \beta_2 v + (1 - \beta_2)(G \odot G)$;

$\hat{m} \leftarrow m / (1 - \beta_1^t)$;

$\hat{v} \leftarrow v / (1 - \beta_2^t)$;

$J_{\mathcal{E}} \leftarrow J_{\mathcal{E}} - \alpha \cdot \hat{m} / (\sqrt{\hat{v}} + \epsilon)$;

$J_{\mathcal{E}} \leftarrow \Pi_{\text{TP}}(\Pi_{\text{PSD}}(J_{\mathcal{E}}))$;

return $J_{\mathcal{E}}$

Apply process
Compute cost
& gradient

ADAM

Updates

Updating Unitary Operators on a Riemannian Manifold

Riemannian Gradient

$$\text{grad } \mathcal{C}(K) = \nabla \mathcal{C}(K) - K, \text{Sym} \left(K^\dagger \nabla \mathcal{C}(K) \right),$$

- Projects the Euclidean gradient onto the tangent space of the unitary manifold.
- Ensures updates respect the manifold geometry.

Cayley Retraction

$$K_{t+1} = \left(I + \frac{\alpha}{2} A_t \right)^{-1} \left(I - \frac{\alpha}{2} A_t \right) K_t,$$

- Retracts updated K back onto the unitary manifold.
- Guarantees that K_{t+1} remains unitary.

Usage

- Use Riemannian gradient for proper direction.
- Apply Cayley retraction after each update to stay on the manifold.

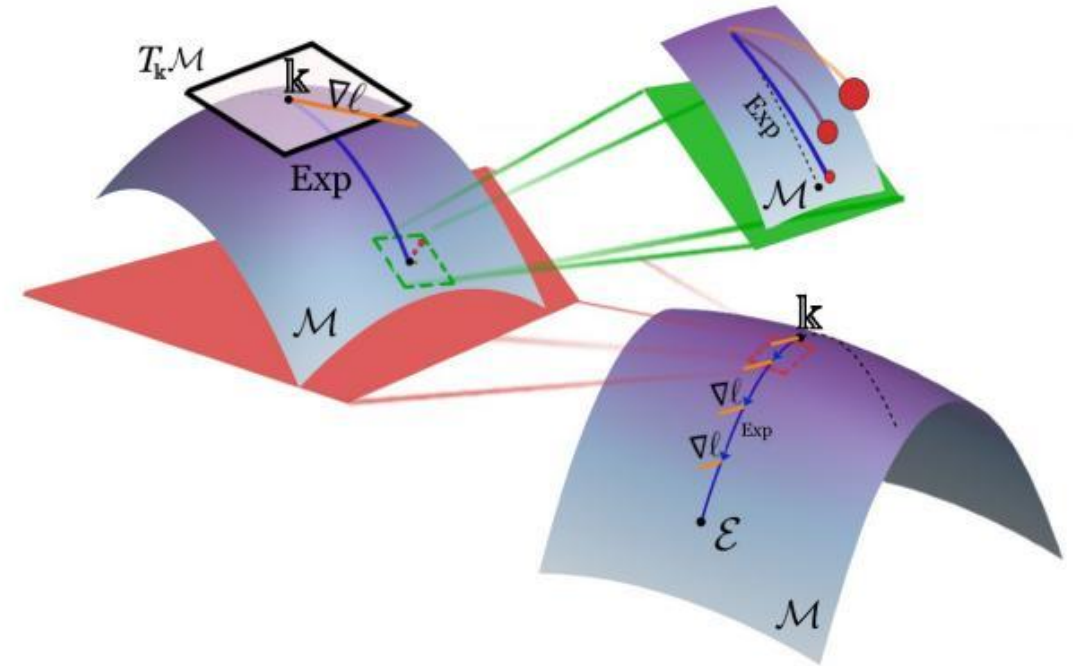


Figure 3.4. Updating K on Riemannian Manifold [5]

3.3. Algorithm on Quantum Circuit

- Prepare a reference state $\rho_0 = |0\rangle\langle 0|$ and a set of random unitaries W_i . Learn the ansatz $V(\theta)$
- Apply the sequence to create full training circuits:

$$\begin{aligned} &\rightarrow V(\theta)(W_i\rho_0W_i^\dagger)V(\theta)^\dagger \\ &\rightarrow U^\dagger[V(\theta)(W_i\rho_0W_i^\dagger)V(\theta)^\dagger]U \\ &\rightarrow W^\dagger[U^\dagger[V(\theta)(W_i\rho_0W_i^\dagger)V(\theta)^\dagger]U]W \end{aligned}$$

Note: This thesis focuses on classical simulation; hybrid execution is planned for future development.

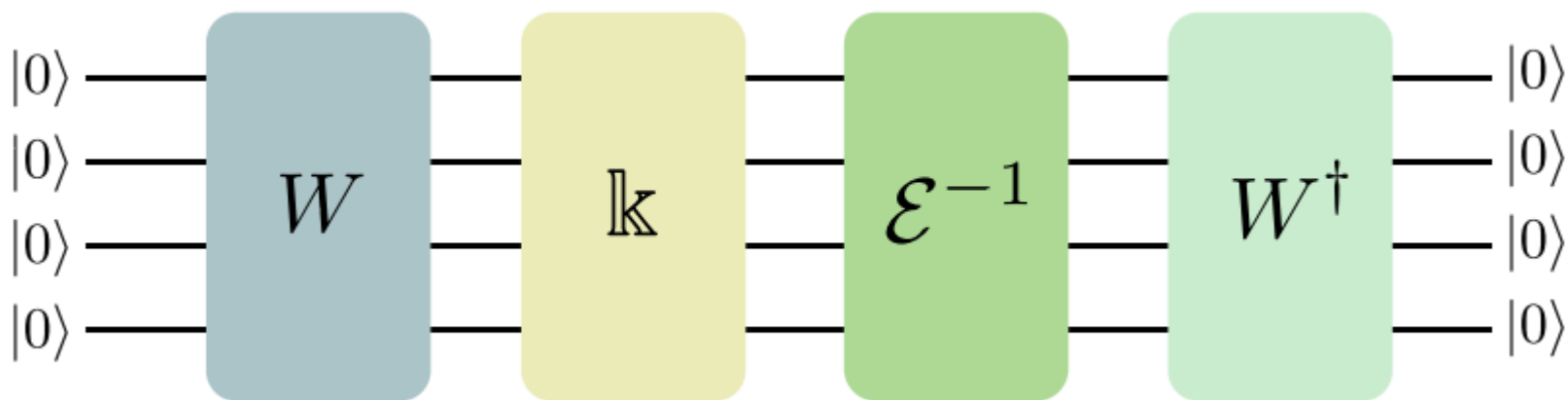


Figure 3.4. Visualization of the circuit sequence

3.3. Algorithm on Quantum Circuit

- The measured probability $p_i(\theta)$ quantifies the similarity between $V(\theta)$ and the target unitary U .
- The cost function is defined as an average over training samples; gradients are estimated via the **parameter-shift rule**, and parameters are updated using **ADAM**.
- When the cost converges to zero, the ansatz $V(\theta)$ successfully learns the target unitary U .

Algorithm 3.3. The implementation of Hybrid CQPT on ansatz

Algorithm 3: Hybrid CQPT on Unitary Ansatz $V(\theta)$

Input: Initial parameters θ_0 , reference state $\rho_0 = |0\rangle\langle 0|$, random unitaries $\{W_i\}_{i=1}^n$, target unitary U , learning rate α , ADAM params $\beta_1, \beta_2, \epsilon$, iterations T , shots N_{shots}

Output: Optimized parameters θ

Initialize $\theta \leftarrow \theta_0, m \leftarrow 0, v \leftarrow 0$;

for $i = 1$ to n do

 Prepare ρ_0 ;

 Apply W_i ;

 Apply ansatz $V(\theta)$;

 Apply inverse target unitary $U^\dagger$;

 Apply $W_i^\dagger$;

Prepare full training circuits
(multiple training states)

for $t = 1$ to T do

$C \leftarrow 0$;

 foreach parameter θ_k do

$\theta_k^+ \leftarrow \theta_k + \pi/2, \theta_k^- \leftarrow \theta_k - \pi/2$;

 Evaluate $C(\theta_k^+)$ and $C(\theta_k^-)$ on pre-built circuits $\{QC_{\text{full}}^{(i)}\}$;

$\frac{\partial C}{\partial \theta_k} \leftarrow \frac{1}{2}[C(\theta_k^+) - C(\theta_k^-)]$;

Use parameter-shift rule for
calculating gradient

$m \leftarrow \beta_1 m + (1 - \beta_1) \nabla C$;

$v \leftarrow \beta_2 v + (1 - \beta_2)(\nabla C \odot \nabla C)$;

$\hat{m} \leftarrow m / (1 - \beta_1^t)$;

$\hat{v} \leftarrow v / (1 - \beta_2^t)$;

$\theta \leftarrow \theta - \alpha \hat{m} / (\sqrt{\hat{v}} + \epsilon)$;

Optimize and Update

return θ

4.1. Benchmarking with Haar random Unitary

Objectives

- Evaluate CQPT accuracy in reconstructing random quantum unitary gates sampled from the Haar distribution.
- Successful reconstruction indicates robustness and general applicability of the CQPT method.

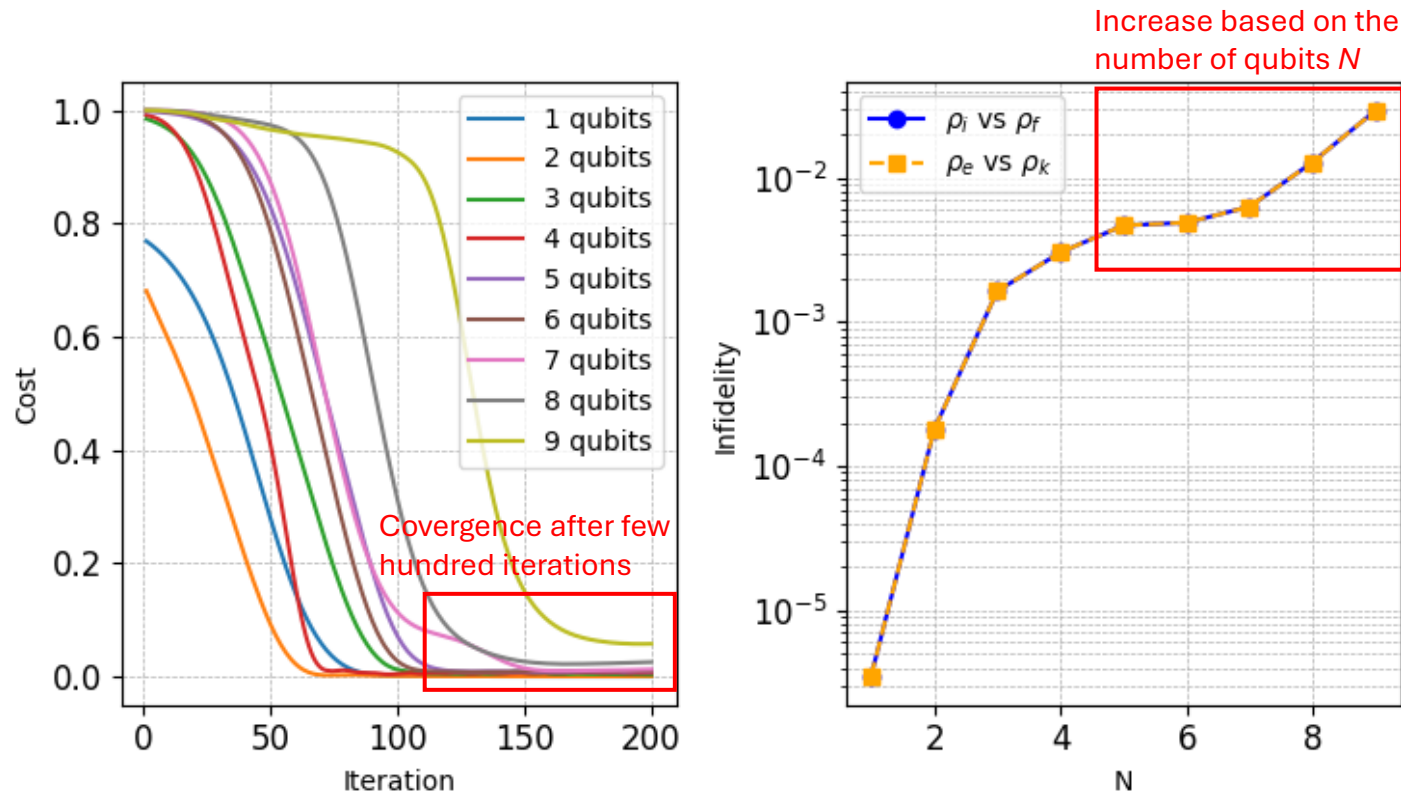
Method

- Target process: a Haar-random unitary $\mathcal{E} \equiv U_{\text{Haar}}$.
- Train a set of Kraus operators $K_1, \dots, K_k$ to approximate the unitary channel.
- Use random input states for training and testing.

Setup

- Numerical simulations on Intel Core i9-10920X, 128 GB RAM, NVIDIA A6000 GPU.

4.1. Benchmarking with Haar random Unitary



```
tomog = KrausTomography()
final_kraus, loss_trace = tomog.fit(
    ...
    gradient_fn=gradient.riemann,
    loss_fn=metric.mean_infidelity,
    ...
)
```

Corresponding code for Fig. 4.1

Figure 4.1. Benchmarking Haar random unitary gates.
(a) Convergence of the cost function $\mathcal{C}(\mathbf{k})$ over iterations.
(b) Average infidelity as a function of the qubit numbers N .

4.1. Benchmarking with Haar random Unitary

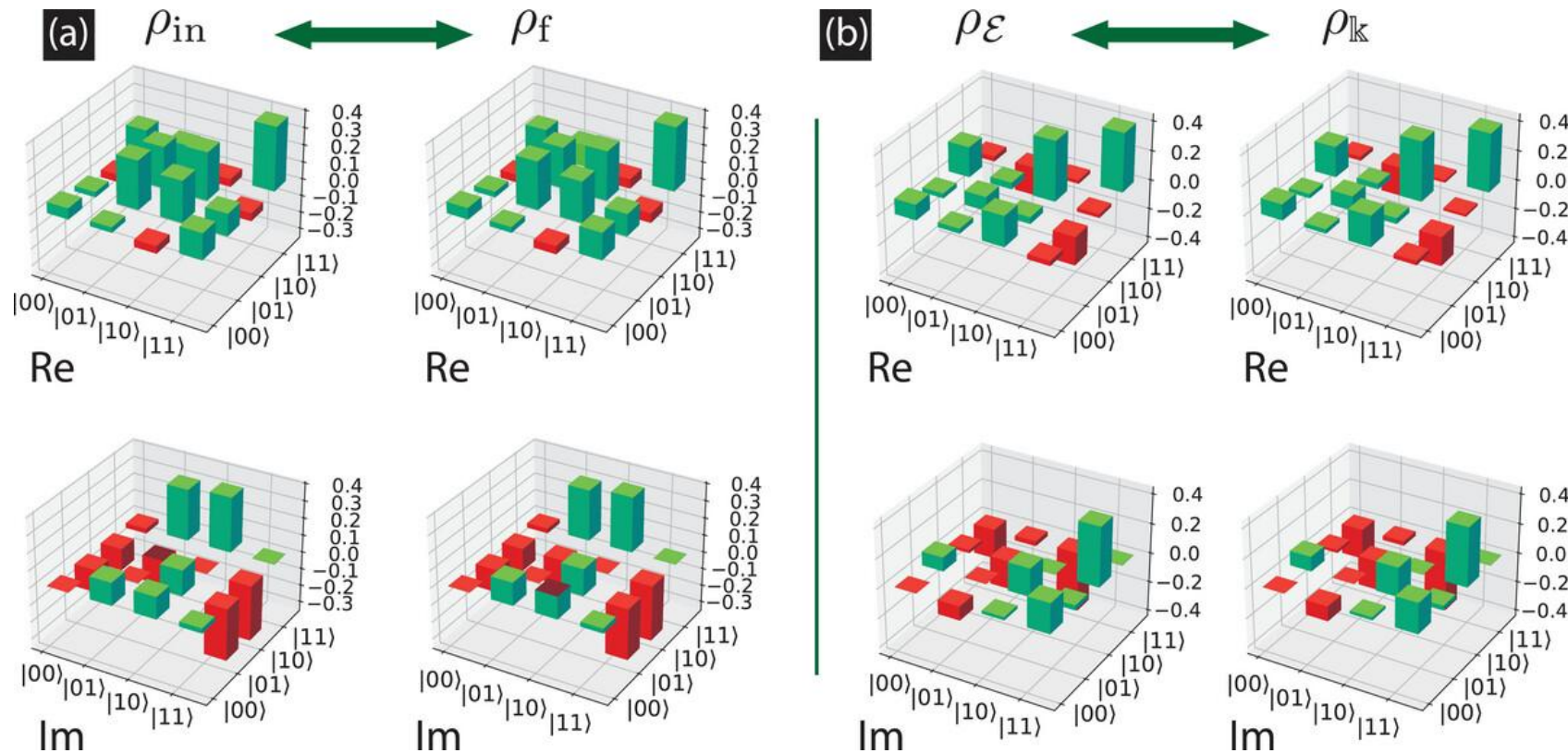


Figure 4.2.

(a) Comparison between the initial and final states, ρ_{in} and ρ_f , where ρ_{in} is selected from the test set. $\Re$ and $\Im$ denote the real and imaginary parts of the matrices, respectively. Green indicates positive values, while red indicates negative values.

(b) Comparison between the intermediate quantum states $\rho_{\mathcal{E}}$ and ρ_k to evaluate the accuracy of the transformation.

4.2. Experiment on Noise models - Dephasing

Objective

- Evaluate Choi-matrix-based CQPT in reconstructing multi-qubit dephasing channels.
- Analyze output-state dependence on the dephasing rate γ . Verify stability and accuracy under varying noise levels.

Dephasing Model

- Single-qubit channel: $\mathcal{E}_j(\rho) = \frac{1+\sqrt{1-\gamma}}{2}\rho + \frac{1-\sqrt{1-\gamma}}{2}\sigma_z\rho\sigma_z$
- Multi-qubit channel: $\mathcal{E}_{total}(\rho) = \prod_{j=1}^N \mathcal{E}_j(\rho)$
- $\gamma = 0$: no noise; $\gamma \rightarrow 1$: maximally mixed state.

Method

- Use Choi-matrix CQPT to reconstruct $\mathcal{E}_{total}$
- Train a variational Choi matrix $J_\mathcal{E}$ with 6^N samples.
- Reconstruction via: $\rho_J = \text{Tr}_x[(\rho_{in}^\top \otimes I)J_\mathcal{E}] \approx \rho_\mathcal{E}$

4.2. Experiment on Noise models - Dephasing

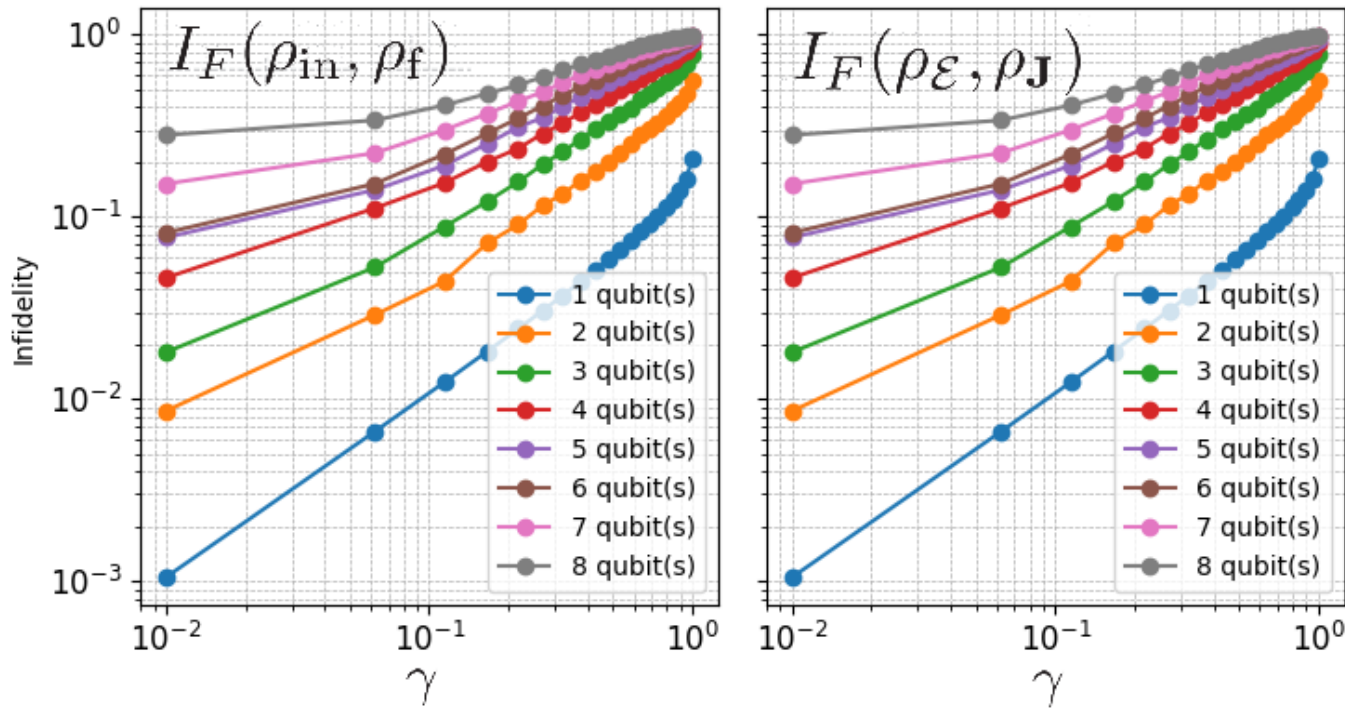


Figure 4.3. CQPT for dephasing noise process.

(a) Average infidelity $I_F(\rho_{in}, \rho_f)$ versus noise γ .

(b) Average infidelity $I_F(\rho_E, \rho_J)$ versus noise γ , showing the effect of noise on optimization performance.

```
tomog = ChoiTomography()
final_matrix, loss_trace = tomog.fit(
    ...
    gradient_fn=gradient.riemann,
    loss_fn=metric.mean_infidelity,
    ...
)
```

Corresponding code for Fig. 4.3

4.2. Experiment on Noise models - Dephasing

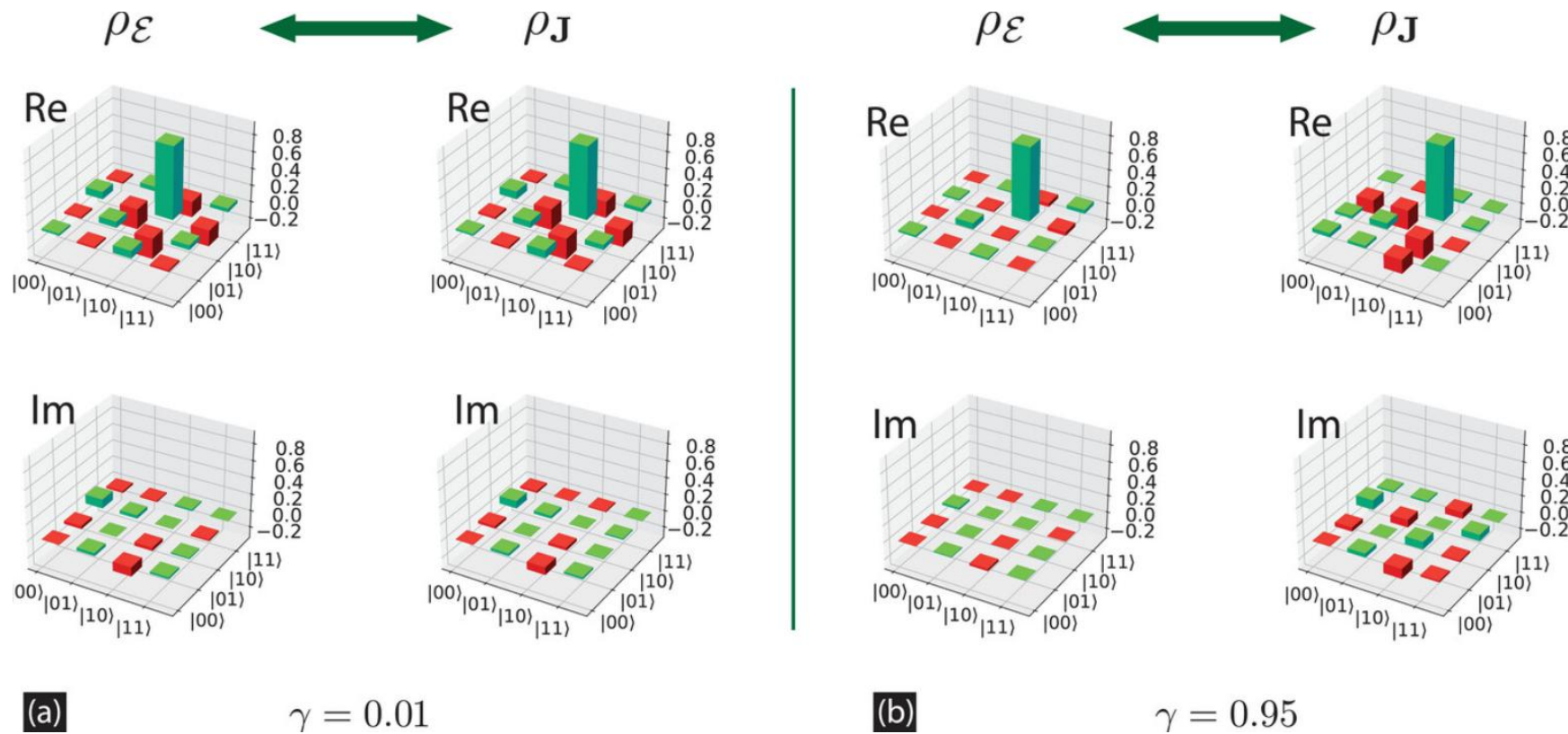


Figure 4.4.

(a) Comparison between the intermediate quantum states ρ_E and ρ_J to evaluate the accuracy of the transformation with $\gamma = 0.01$ (small noise).

(b) Comparison between the intermediate quantum states ρ_E and ρ_J to evaluate the accuracy of the transformation with $\gamma = 0.95$ (large noise).

4.2. Experiment on Noise models – Depolarizing & Amplitude-damping

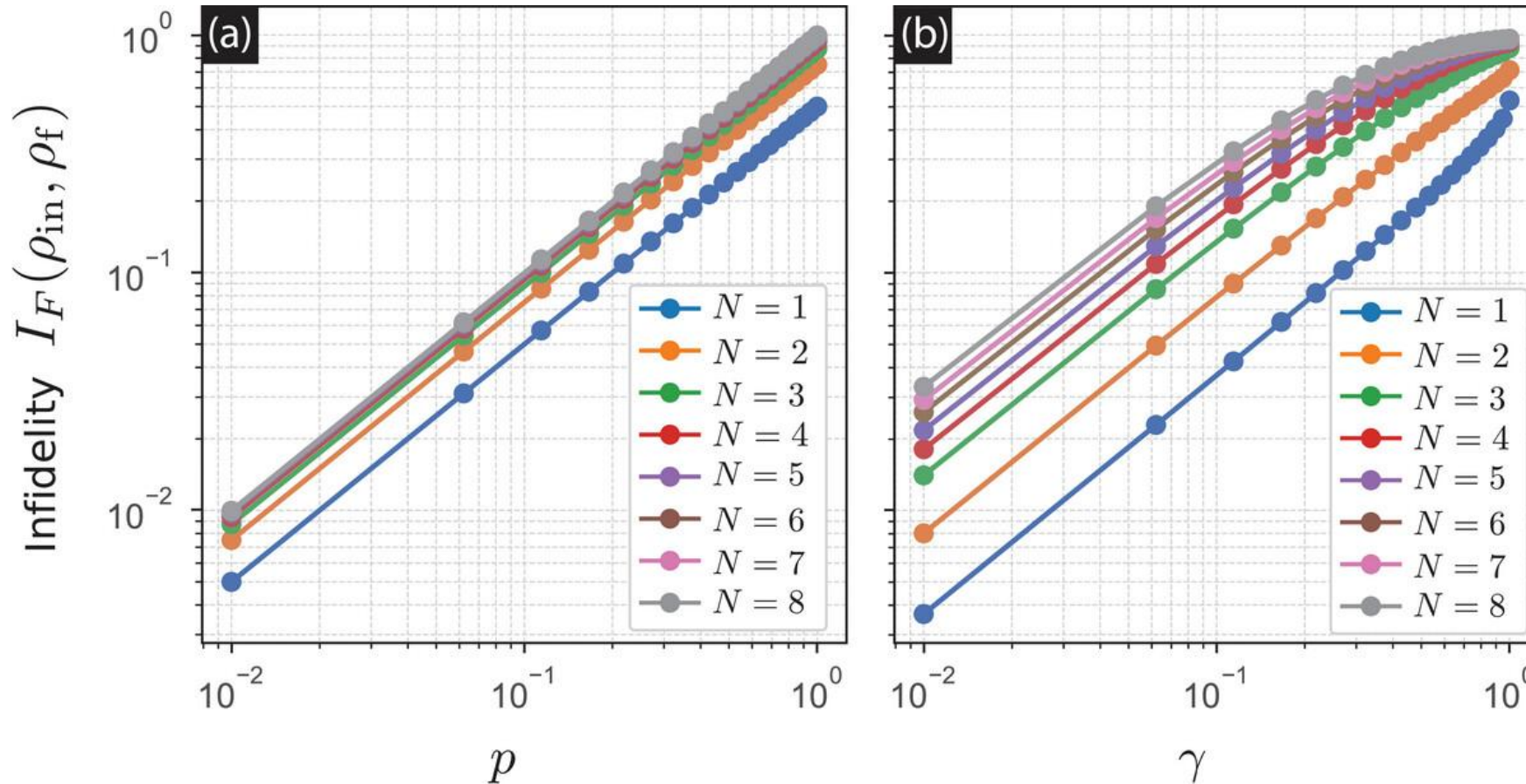


Figure 4.5.

CQPT for depolarizing and amplitude-damping process.

(a) Average infidelity vs. the depolarizing strength.

(b) Average infidelity vs. the amplitude-damping rate .

4.3. Comparison

- The proposed method is compared with prior works [5][6] using Riemannian gradient-based QPT.
- Previous approaches (Measurement-based QPT, MQPT) directly compare measurement outcomes of $\rho_{\mathcal{E}}$ and ρ_k using a cost function over many POVM elements.

MQPT

- Uses $n = 3^N$ input states and $p = 3^N$ measurements per state.
- Total measurements per iteration: $3^N \times 3^N$.
- Measurement time: $T_{\text{MQPT}} = c_1 \times 3^{2N}$

Proposed CQPT (Compilation-based)

- Applies the channels directly and uses **one measurement per state**.
- Total measurements per iteration: 3^N .
- Measurement time: $T_{\text{CQPT}} = c_2 \times 3^N$

[5] Ahmed S., Quijandría F., Kockum A. F., “Gradient-descent quantum process tomography,” Phys. Rev. Lett., 2023.

[6] Volya D., Nikitin A., Mishra P., “Fast quantum process tomography via riemannian gradient descent,” 2024.

4.3. Comparison

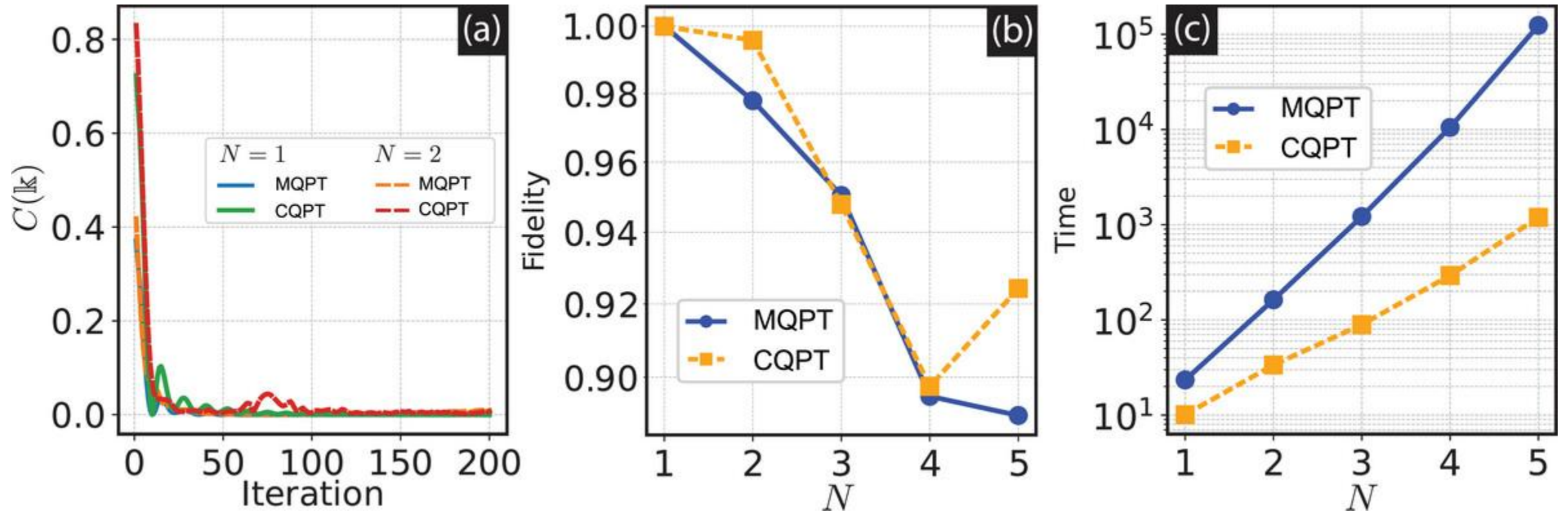


Figure 4.6. Comparison results of MQPT and CQPT.

(a) Cost function $C(\mathbf{k})$ versus the number of iterations for systems with $N = 1$ and $N = 2$ qubits.

(b) Plot of the fidelity as a function of the number of qubits N for both methods.

(b) Execution time as a function of the number of qubits N .

4.4. Initial experiment on Quantum circuit

Objectives

- Validate the feasibility of CQPT in a quantum-circuit setting using a hybrid quantum–classical approach.
- Unlike previous purely numerical simulations, this experiment:
 - Executes **full training on quantum circuits** (state preparation, parameterized gates, measurement).
 - Uses classical optimization to update parameters.
 - Serves as a **proof-of-concept** for future deployment on real quantum hardware.

System

- $N = 2$ qubits.
- Target operator: Random unitary circuit U_{target} (depth = 15).
- Training data: $3^N = 9$ random circuits W_i (depth = 10).
- Ansatz:
 - Reversible W-chain circuit with ZXZ rotations, 3 layers.
 - Parameters $\theta \sim [0, 2\pi]$ (random initialization).

4.4. Initial experiment on Quantum circuit

Training circuit structure

$$W_i \rightarrow V(\theta) \rightarrow U_{\text{target}}^\dagger \rightarrow W_i^\dagger$$

- If $V(\theta) = U_{\text{target}}$, the final state returns to $|0\rangle^{\otimes N}$.

Training Procedure

- Simulation backend: Qiskit StatevectorSampler (noise-free, ideal simulation).
- Measurements:
 - Computational basis, 100 shots per circuit.
- Cost function: $\mathcal{C}(\theta) = 1 - \frac{1}{n} \sum_{i=1}^n p_0^{(i)}(\theta)$
 - $p_0^{(i)}$: probability of measuring $|0\rangle^{\otimes N}$

Optimization

- Gradients estimated via parameter-shift rule $\left(\pm \frac{\pi}{2}\right)$.
- Parameters updated using ADAM optimizer.

4.4. Initial experiment on Quantum circuit

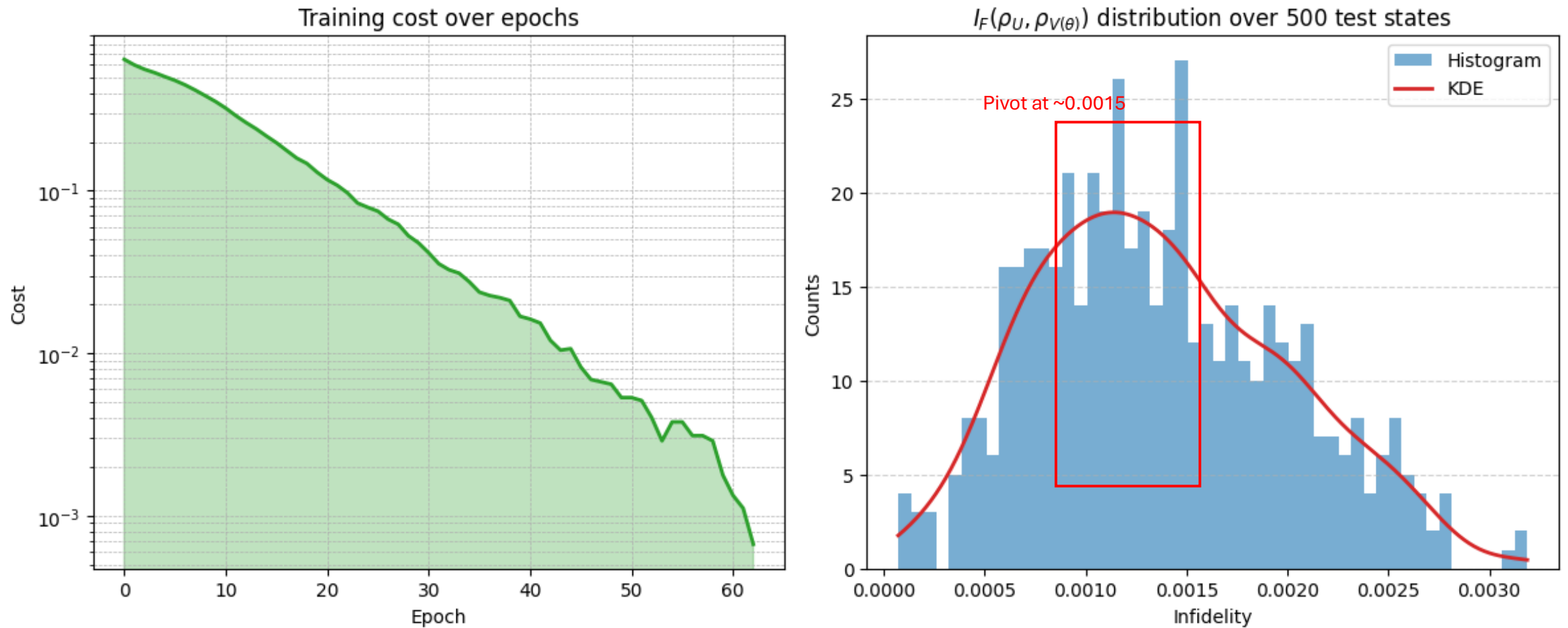


Figure 4.7. Cost function values and the distribution of $I_F(\rho_U, \rho_{V(\theta)})$ over 500 test states.

4.4. Initial experiment on Quantum circuit

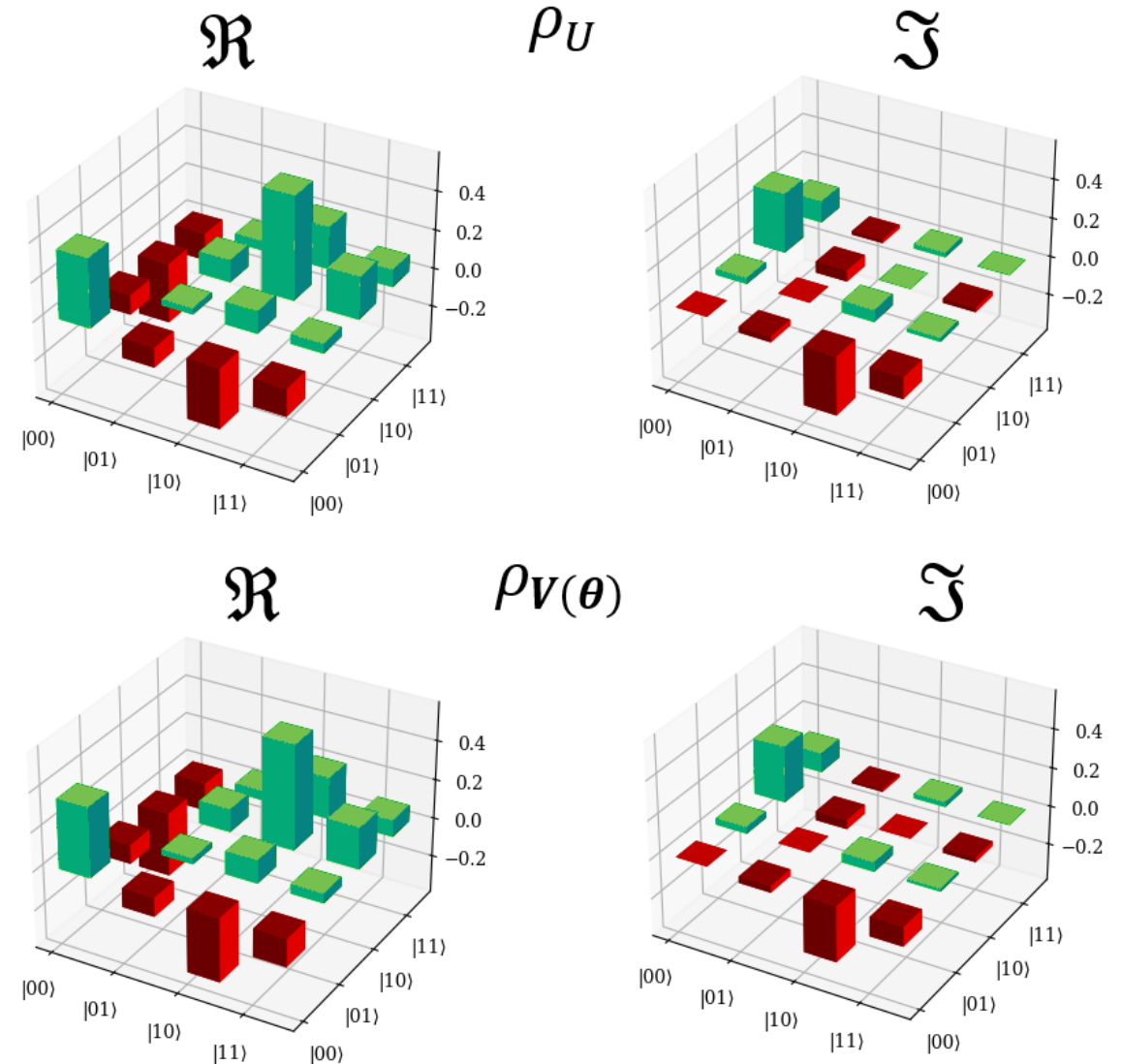
Infidelity distribution on test states

- Most values range $\sim 0.0001 - 0.003$, peak ~ 0.0015 .
- Narrow distribution with no large outliers
→ good generalization beyond training set.

Visual comparison shows high similarity.

- Reversible ansatz $V(\theta)$ accurately approximates random unitary operators.
- Validates the feasibility and effectiveness of CQPT.
- Provides a foundation for extending to larger qubit systems and real quantum hardware.

Figure 4.8. Comparison between ρ_U and $\rho_{V(\theta)}$, with the input state ρ chosen randomly.



Appendix

Journal paper

Huynh Le Dan Linh, Vu Tuan Hai, Le Bin Ho, “Advancing quantum process tomography through quantum compilation”, in Advanced Quantum Technologies, 2026 <https://doi.org/10.1002/qute.202500494>